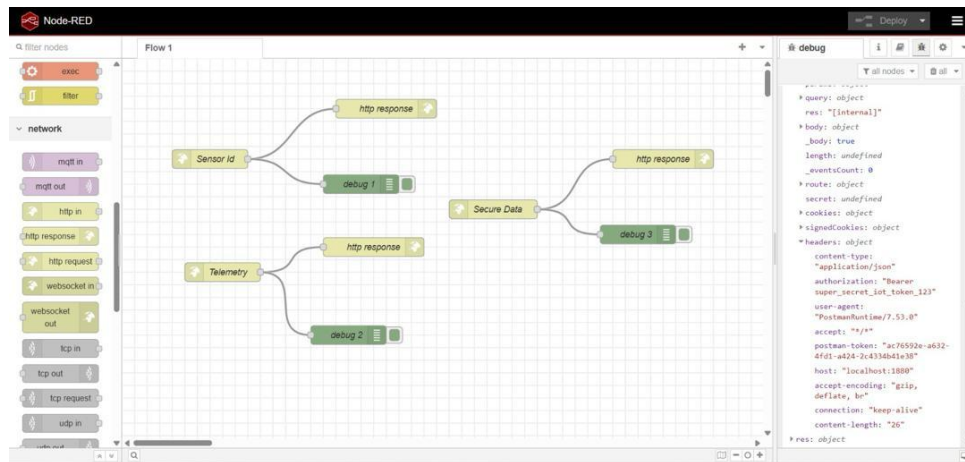
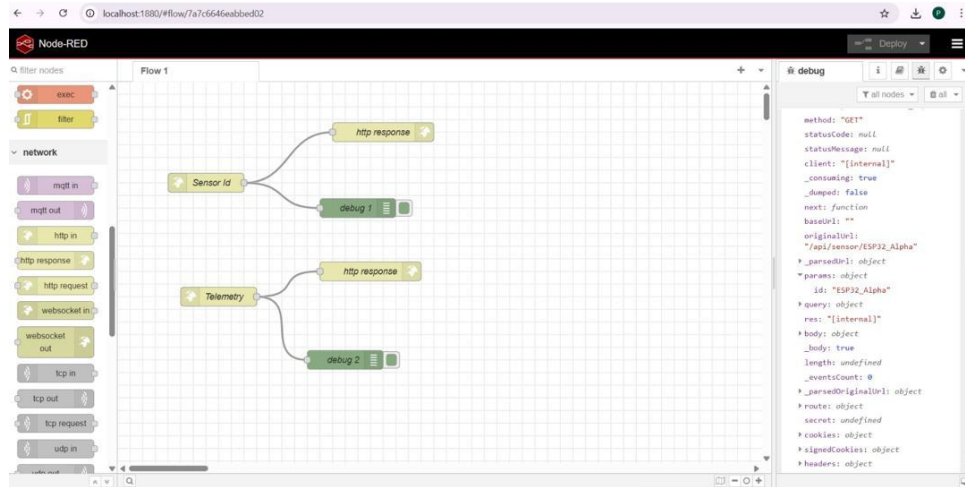
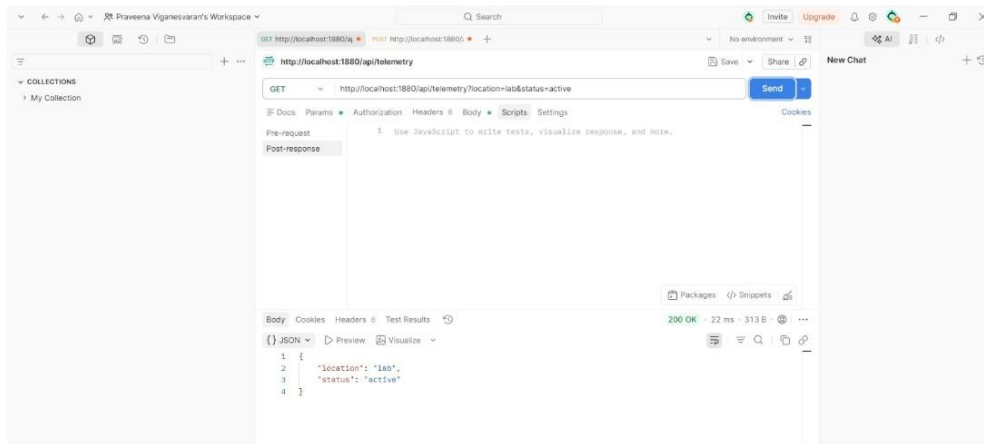


## EXERCISE 3: PACKET INSPECTION

### Step 1: Query Strings vs. Route Parameters in Node-RED

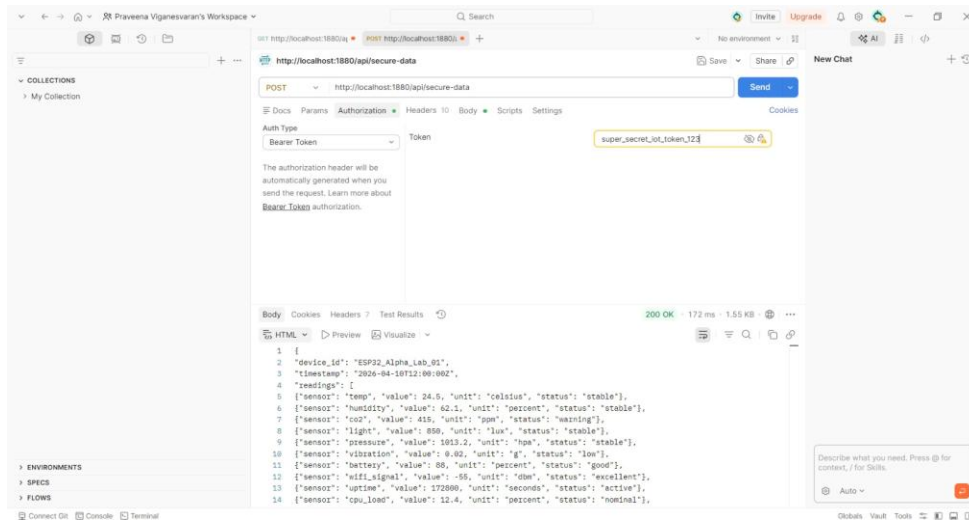


(Node-RED)



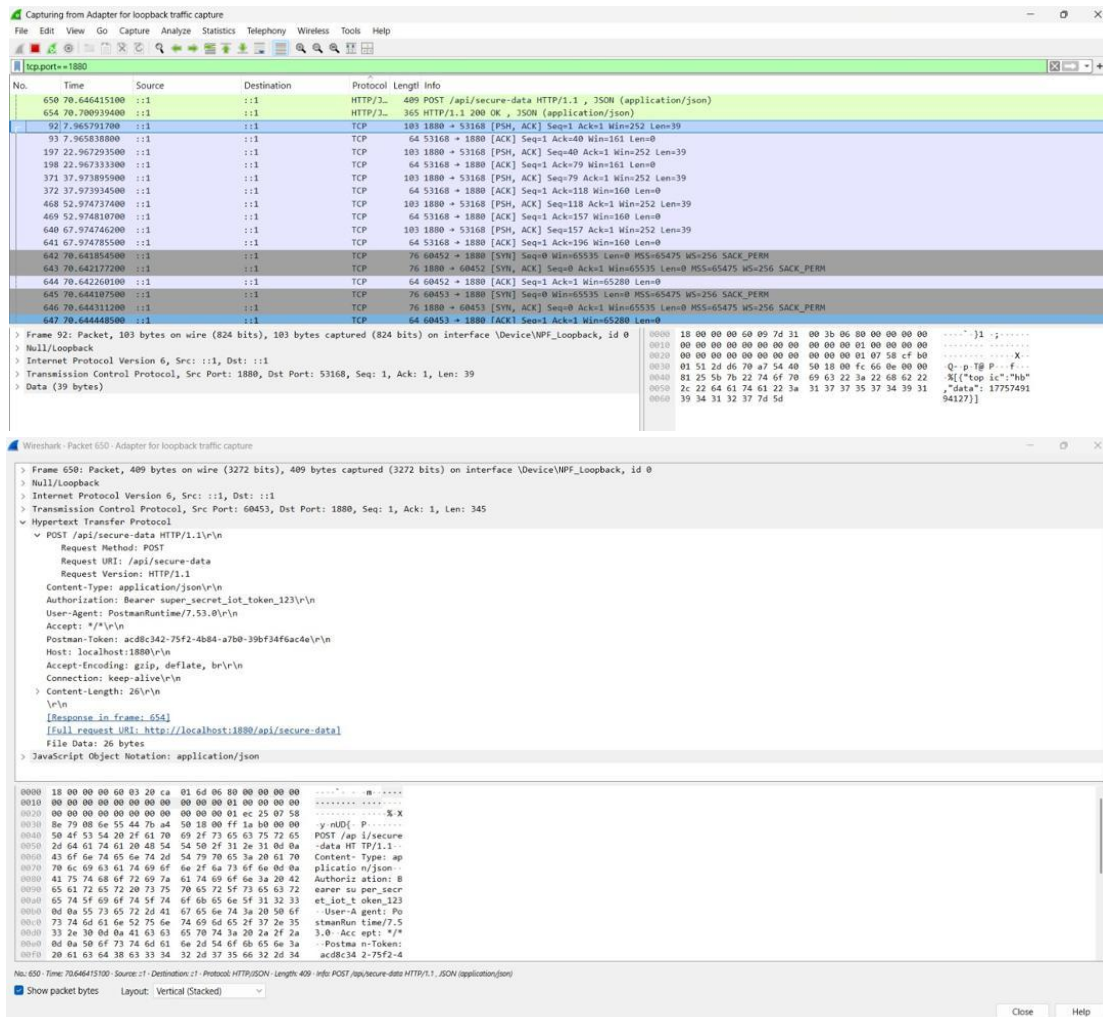
(Postman)

## Step 2: The Power of HTTP Headers (Media & Authentication)



(Postman)

## Step 3: Deep Packet Inspection with Wireshark



## Part 4: Analytical Modifications & Network Comparison

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port=1880

| No.  | Time        | Source | Destination | Protocol | Length | Info                                                                                       |
|------|-------------|--------|-------------|----------|--------|--------------------------------------------------------------------------------------------|
| 3148 | 260.4978062 | :::1   | :::1        | HTTP     | 397    | POST /api/secure-data HTTP/1.1 (text/plain)                                                |
| 3164 | 260.3239392 | :::1   | :::1        | HTTP     | 321    | HTTP/1.1 200 OK (text/html)                                                                |
| 5181 | 5.07393400  | :::1   | :::1        | HTTP     | 1724   | POST /api/secure-data HTTP/1.1 (text/plain)                                                |
| 5185 | 5.056585100 | :::1   | :::1        | HTTP     | 1640   | HTTP/1.1 200 OK (text/html)                                                                |
| 650  | 544.7799662 | :::1   | :::1        | HTTP/2.  | 409    | POST /api/secure-data HTTP/1.1, 350N (application/json)                                    |
| 654  | 544.7254719 | :::1   | :::1        | HTTP/2.  | 365    | HTTP/1.1 200 OK, 350N (application/json)                                                   |
| 92   | 607.4606196 | :::1   | :::1        | TCP      | 103    | 1880 → 53168 [PSH, ACK] Seq=1 Ack=1 Win=252 Len=39                                         |
| 93   | 607.4606725 | :::1   | :::1        | TCP      | 64     | 53168 → 1880 [ACK] Seq=1 Ack=40 Win=161 Len=0                                              |
| 197  | 592.4591178 | :::1   | :::1        | TCP      | 103    | 1880 → 53168 [PSH, ACK] Seq=40 Ack=1 Win=252 Len=39                                        |
| 198  | 592.4590780 | :::1   | :::1        | TCP      | 64     | 53168 → 1880 [ACK] Seq=1 Ack=79 Win=161 Len=0                                              |
| 371  | 577.4525154 | :::1   | :::1        | TCP      | 103    | 1880 → 53168 [PSH, ACK] Seq=79 Ack=1 Win=252 Len=39                                        |
| 372  | 577.4524768 | :::1   | :::1        | TCP      | 64     | 53168 → 1880 [ACK] Seq=118 Ack=1 Win=160 Len=0                                             |
| 468  | 562.4516739 | :::1   | :::1        | TCP      | 103    | 1880 → 53168 [PSH, ACK] Seq=118 Ack=1 Win=252 Len=39                                       |
| 469  | 562.4516006 | :::1   | :::1        | TCP      | 64     | 53168 → 1880 [ACK] Seq=1 Ack=157 Win=160 Len=0                                             |
| 640  | 547.4516651 | :::1   | :::1        | TCP      | 103    | 1880 → 53168 [PSH, ACK] Seq=157 Ack=1 Win=252 Len=39                                       |
| 641  | 547.4516258 | :::1   | :::1        | TCP      | 64     | 53168 → 1880 [ACK] Seq=1 Ack=196 Win=160 Len=0                                             |
| 642  | 544.7845568 | :::1   | :::1        | TCP      | 76     | 60452 → 1880 [SYN] Seq=0 Win=0 Len=0 MSS=65535 Len=0 MSS=65475 WS=256 SACK_PERM            |
| 643  | 544.7845241 | :::1   | :::1        | TCP      | 76     | 1880 → 60452 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0 MSS=65535 Len=0 MSS=65475 WS=256 SACK_PERM |

Frame 3148: Packet, 397 bytes on wire (3176 bits), 397 bytes captured (3176 bits) on interface \Device\NPF\_{...}\_Loopback

Null/Loopback

Internet Protocol Version 6, Src: :::1, Dst: :::1

Transmission Control Protocol, Src Port: 62758, Dst Port: 1880, Seq: 1, Ack: 1, Len: 333

Hypertext Transfer Protocol

POST /api/secure-data HTTP/1.1\r\n

Request Method: POST

Request URI: /api/secure-data

Request Version: HTTP/1.1

Content-Type: text/plain\r\n

Authorization: Bearer super\_secret\_123\r\n

User-Agent: PostmanRuntime/7.53.0\r\n

Accept: \*/\*\r\n

Postman-Token: 585ac558-d1e5-4d45-bc9c-ba22c63bb5c5\r\n

Host: localhost:1880\r\n

Accept-Encoding: gzip, deflate, br\r\n

Connection: keep-alive\r\n

Content-Length: 20\r\n

Wireshark · Packet 5181 · Adapter for loopback traffic capture

> Frame 5181: Packet, 1724 bytes on wire (13792 bits), 1724 bytes captured (13792 bits) on interface \Device\NPF\_{...}\_Loopback, id 0

> Null/Loopback

> Internet Protocol Version 6, Src: :::1, Dst: :::1

> Transmission Control Protocol, Src Port: 62231, Dst Port: 1880, Seq: 1, Ack: 1, Len: 1660

> Hypertext Transfer Protocol

POST /api/secure-data HTTP/1.1\r\n

Request Method: POST

Request URI: /api/secure-data

Request Version: HTTP/1.1

Content-Type: text/plain\r\n

Authorization: Bearer super\_secret\_123\r\n

User-Agent: PostmanRuntime/7.53.0\r\n

Accept: \*/\*\r\n

Postman-Token: 16185442-e306-44c0-8491-300c349a3675\r\n

Host: localhost:1880\r\n

Accept-Encoding: gzip, deflate, br\r\n

Connection: keep-alive\r\n

Content-Length: 1345\r\n

\r\n

[Response in frame: 5185]

[Full request URI: http://localhost:1880/api/secure-data]

File Data: 1345 bytes

> Line-based text data: text/plain (25 lines)

```

0000 18 00 00 00 60 03 7e 59 06 90 06 80 00 00 00 00  .....Y.....
0010 00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00  .....
0020 00 00 00 00 00 00 00 00 00 00 00 01 f3 17 07 58  .....X
0030 6a e9 c7 49 5d 90 58 4f 50 18 00 ff 8e ae 00 00  .....j-..XO P-----
0040 50 4f 53 54 20 2f 61 70 69 2f 73 65 63 75 72 65  POST /ap i/secure
0050 2d 64 61 74 61 20 48 54 54 50 2f 31 2e 31 0d 0a  -data HT TP/1.1-
0060 43 6f 6e 74 65 6e 74 2d 54 70 70 65 3a 20 74 65  Content- Type: te
0070 78 74 2f 70 6c 61 69 6e 0d 0a 41 75 74 68 6f 72  xt/plain - Author
0080 69 7a 61 74 69 6f 6e 3a 20 42 65 61 72 65 72 20  ization: Bearer
0090 73 75 70 65 72 5f 73 65 63 72 65 74 5f 69 6f 74  super_se cret_1ot
00a0 5f 74 6f 6b 65 6e 5f 31 32 33 0d 0a 55 73 65 72  _token_1 23- User
00b0 2d 41 67 65 6e 74 20 50 6f 73 74 6d 61 6e 52  -Agent: PostmanR
00c0 75 6e 74 69 6d 65 2f 37 2e 35 33 2e 30 0d 0a 41  untime/7. 53.0 -A
00d0 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 50 6f 73 74  cept: */*-Post
00e0 6d 61 6e 2d 54 6f 6b 65 6e 3a 20 31 36 31 38 35  man-Toke n: 16185
00f0 34 34 32 2d 65 33 30 36 2d 34 34 63 30 2d 38 34  442-e306 -44c0-84
0100 39 31 2d 33 30 30 63 33 34 39 61 33 36 37 35 0d  91-300c3 49a3675

```

No: 5181 · Time: 5.07393400 · Source: :::1 · Destination: :::1 · Protocol: HTTP · Length: 1724 · Info: POST /api/secure-data HTTP/1.1 (text/plain)

Show packet bytes Layout: Vertical (Stacked)



Wireshark · Packet 650 · Adapter for loopback traffic capture

> Frame 650: Packet, 409 bytes on wire (3272 bits), 409 bytes captured (3272 bits) on interface \Device\NPF\_{Loopback}, id 0

> Null/Loopback

> Internet Protocol Version 6, Src: ::1, Dst: ::1

> Transmission Control Protocol, Src Port: 60453, Dst Port: 1880, Seq: 1, Ack: 1, Len: 345

> Hypertext Transfer Protocol

> POST /api/secure-data HTTP/1.1\r\n

Request Method: POST

Request URI: /api/secure-data

Request Version: HTTP/1.1

Content-Type: application/json\r\n

Authorization: Bearer super\_secret\_iot\_token\_123\r\n

User-Agent: PostmanRuntime/7.53.0\r\n

Accept: \*/\*\r\n

Postman-Token: acd8c342-75f2-4b84-a7b0-39bf34f6ac4e\r\n

Host: localhost:1880\r\n

Accept-Encoding: gzip, deflate, br\r\n

Connection: keep-alive\r\n

> Content-Length: 26\r\n\r\n

0030 8e 79 08 6e 55 44 7b a4 50 18 00 ff 1a b0 00 00 -y-nUD{ P-.....

0040 50 4f 53 54 20 2f 61 70 69 2f 73 65 63 75 72 65 POST /ap i/secure

0050 2d 64 61 74 61 20 48 54 54 50 2f 31 2e 31 0d 0a -data HT TP/1.1--

0060 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 61 70 Content- Type: ap

0070 70 6e 69 63 61 74 69 6f 6e 2f 6a 73 6f 6e 0d 0a plicatio n/json--

0080 41 75 74 68 6f 72 69 7a 61 74 69 6f 6e 3a 20 42 Authoriz ation: B

0090 65 61 72 65 72 20 73 75 70 65 72 5f 73 65 63 72 earer su per\_secr

00a0 65 74 5f 69 6f 74 5f 74 6f 6b 65 6e 5f 31 32 33 et\_iot\_t oken\_123

00b0 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 50 6f ..User-A gent: Po

00c0 73 74 6d 61 6e 52 75 6e 74 69 6d 65 2f 37 2e 35 stmanRun time/7.5

00d0 33 2e 30 0d 0a 41 63 63 65 70 74 3a 20 2a 2f 2a 3.0..Acc ept: \*/\*

00e0 0d 0a 50 6f 73 74 6d 61 6e 2d 54 6f 6b 65 6e 3a ..Postma n-Token:

00f0 20 61 63 64 38 63 33 34 32 2d 37 35 66 32 2d 34 acd8c34 2-75f2-4

0100 62 38 34 2d 61 37 62 30 2d 33 39 62 66 33 34 66 b84-a7b0 -39bf34f

0110 36 61 63 34 65 0d 0a 48 6f 73 74 3a 20 6c 6f 63 6ac4e..H ost: loc

0120 61 6c 68 6f 73 74 3a 31 38 38 30 0d 0a 41 63 63 alhost:1 880..Acc

0130 65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 7a ept-Enco ding: gz

0140 69 70 2c 20 64 65 66 6c 61 74 65 2c 20 62 72 0d ip, defl ate, br.

0150 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 -Connect ion: kee

0160 70 2d 61 6c 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 p-alive..Content

0170 2d 4c 65 6e 67 74 68 3a 20 32 36 0d 0a 0d 0a 20 -length: 26....

0180 7b 22 73 74 61 74 75 73 22 3a 20 22 73 65 63 75 {"status": "secu

0190 72 65 20 74 65 73 74 22 7d re test" }

HTTP Content-Type header (http.content\_type), 32 bytes

☒ Show packet bytes Layout: Vertical (Stacked)

Wireshark · Packet 5181 · Adapter for loopback traffic capture

> Transmission Control Protocol, Src Port: 62231, Dst Port: 1880, Seq: 1, Ack: 1, Len: 1660

> Hypertext Transfer Protocol

> POST /api/secure-data HTTP/1.1\r\n

Request Method: POST

Request URI: /api/secure-data

Request Version: HTTP/1.1

Content-Type: text/plain\r\n

Authorization: Bearer super\_secret\_iot\_token\_123\r\n

User-Agent: PostmanRuntime/7.53.0\r\n

Accept: \*/\*\r\n

Postman-Token: 16185442-e306-44c0-8491-300c349a3675\r\n

Host: localhost:1880\r\n

Accept-Encoding: gzip, deflate, br\r\n

Connection: keep-alive\r\n

> Content-Length: 1345\r\n\r\n

[Response in frame: 5185]

[Full request URI: http://localhost:1880/api/secure-data]

File Data: 1345 bytes

0060 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 Content- Type: te

0070 78 74 2f 70 6c 61 69 6e 0d 0a 41 75 74 68 6f 72 xt/plain..Author

0080 69 7a 61 74 69 6f 6e 3a 20 42 65 61 72 65 72 20 ization: Bearer

0090 73 75 70 65 72 5f 73 65 63 72 65 74 5f 69 6f 74 super\_se cret\_iot

00a0 5f 74 6f 6b 65 6e 5f 31 32 33 0d 0a 55 73 65 72 \_token\_1 23..User

00b0 2d 41 67 65 6e 74 3a 20 50 6f 73 74 6d 61 6e 52 -Agent: PostmanR

00c0 75 6e 74 69 6d 65 2f 37 2e 35 33 2e 30 0d 0a 41 untime/7 .53.0..A

00d0 63 65 70 74 3a 20 2a 2f 2a 0d 0a 50 6f 73 74 cept: \* /.\*..Post

00e0 6d 61 6e 2d 54 6f 6b 65 6e 3a 20 31 36 31 38 35 man-Toke n: 16185

00f0 34 34 32 2d 65 33 30 36 2d 34 34 63 30 2d 38 34 442-e306 -44c0-84

0100 39 31 2d 33 30 30 63 33 34 39 61 33 36 37 35 0d 91-300c3 49a3675.

0110 0a 48 6f 73 74 3a 20 6c 6f 63 61 6c 68 6f 73 74 .Host: l ocalhost

0120 3a 31 38 38 30 0d 0a 41 63 63 65 70 74 2d 45 6e :1880..A ccept-En

0130 63 6f 64 69 6e 67 3a 20 67 7a 69 70 2c 20 64 65 coding: gzip, de

0140 66 6c 61 74 65 2c 20 62 72 0d 0a 43 6f 6e 6e 65 flate, b r..Conne

0150 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 ction: k eep-aliv

0160 65 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74 e..Conte nt-Lengt

0170 68 3a 20 31 33 34 35 0d 0a 0d 0a 7b 0d 0a 20 20 h: 1345. ...{..

0180 22 64 65 76 69 63 65 5f 69 64 22 3a 20 22 45 53 "device\_id": "ES

0190 50 33 32 5f 41 6c 70 68 61 5f 4c 61 62 5f 30 31 P32\_Alph a\_Lab\_01

01a0 22 2c 0d 0a 20 20 22 74 69 6d 65 73 74 61 6d 70 ",.. "t imestamp

01b0 22 3a 20 22 32 30 32 36 2d 30 34 2d 31 30 54 31 "; "2026 -04-10T1

01c0 32 3a 30 30 3a 30 30 5a 22 2c 0d 0a 20 20 22 72 2:00:00Z ",.. "r

HTTP Content-Type header (http.content\_type), 26 bytes

☒ Show packet bytes Layout: Vertical (Stacked)

(Wireshark)